

Telecom Customer Churn Prediction - Final Report

1. Dataset Overview

- Total customers: 7043
- Features: 21
- Churn rate: 26.5%

2. Model Performance

Best Model: Gradient Boosting Classifier

- Accuracy: 79.4%
- Precision: 64.6%
- Recall: 49.7%
- F1-Score: 0.562
- ROC-AUC: 0.841

3. Key Technical Fix

Used max_depth=3 for Gradient Boosting to prevent overfitting on 1407 test samples.
This improved Recall from 0.30 to 0.497.

4. Business Output - Risk Tiers

Low Risk: 858 customers

Medium Risk: 368 customers

High Risk: 181 customers

5. Recommendation

Target 181 High-risk customers immediately with retention offers.

They have >60% probability of churning.

Conclusion: Model identifies 49.7% of actual churners with 84.1% AUC. Ready for deployment.